

Band-Limitedness of the Hardy Z-Function in the Phase Variable

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Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, and $\theta(t) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta function. Set $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$; for any constant $c > c_*$, define the shifted phase $\Theta(t) := \theta(t) + ct$. Then $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection. The unitary inverse phase pullback

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}, \quad (1)$$

with the principal (positive) square root of the strictly positive real quantity $\Theta'(\Theta^{-1}(u))$, is a real-valued function on $\mathbb{R}$ whose Fourier support is contained in $[-1, 1]$. By the Paley–Wiener theorem, Y extends to an entire function of exponential type ≤ 1 . Non-negativity of the spectral measure $S := |\widehat{Y}|^2$ together with Akhiezer’s theorem places Y in the Laguerre–Pólya class, so all complex zeros of Y are real. On $\mathbb{R}$ the identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is the rearrangement of (1), with $\sqrt{\Theta'(t)} > 0$. This unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$ is a bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$ with preservation of multiplicity. The critical-line identity $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ for $t \in \mathbb{R}$, combined with reality of zeros of Y and the unitary transfer, fixes the zeros of ζ on the critical line as the Θ^{-1} -image of the real zeros of Y .

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-valued on $\mathbb{R}$. The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$ ties the zeros of Z on $\mathbb{R}$ to the zeros of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$ under $s = \frac{1}{2} + it$: the map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$, and $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$ makes the two sides vanish simultaneously.

The Riemann–Siegel phase θ fails to be monotone on $[0, \infty)$: θ' is strictly increasing on $[0, \infty)$ but negative on $[0, t_c]$ and positive on $[t_c, \infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ' . Monotonicity on all of $\mathbb{R}$ is restored by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_* := -\theta'(0)$, which enforces $\Theta' > 0$ on $\mathbb{R}$ and makes $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ a C^∞ bijection. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is defined in terms of this real bijection.

All spectral quantities, in particular the mode thresholds β_n , are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y : $\operatorname{supp} S \subseteq [-1, 1]$. Paley–Wiener then gives Y entire of exponential type ≤ 1 ; Akhiezer gives reality of its complex zeros. The real-line identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is a unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$, and so is a zero-set bijection with multiplicity preservation between Y on $\mathbb{R}$ and Z

on $\mathbb{R}$ (Theorem 3). The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ on $t \in \mathbb{R}$ yields the conclusion for the zeros of ζ on the critical line.

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (2)$$

$$\theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (3)$$

$$\theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right) \quad (4)$$

For $t \in \mathbb{R}$, $\frac{1}{4} + \frac{it}{2}$ avoids $\mathbb{Z}_{\leq 0}$ (its real part is $\frac{1}{4}$), so $\theta, \theta', \theta''$ are C^∞ on $\mathbb{R}$.

Lemma 1 (Positivity of θ''). *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \quad (k \geq 0, t > 0) \quad (5)$$

Summing, $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. $\square$

Lemma 2 (Exact value of $\theta'(0)$).

$$\theta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right) \quad (6)$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$ ([1], §6.3.4). $\square$

Definition 1 (Admissible constant).

$$c_* := -\theta'(0) = \frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right). \quad (7)$$

A real c is admissible if $c > c_*$. Throughout the paper c is admissible.

Lemma 3 (Monotone bijection). *For every admissible c , $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection.*

Proof. θ' is even on $\mathbb{R}$ (from (3); $\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2})$ is even in τ) and by Lemma 1 $\theta'' > 0$ on $(0, \infty)$, so θ' is strictly increasing on $[0, \infty)$ and attains its minimum on $\mathbb{R}$ at $\tau = 0$, namely $\theta'(0) = -c_*$. For any $c > c_*$, $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$ on $\mathbb{R}$. The standard asymptotic $\theta(\tau) = \frac{\tau}{2} \log(\tau/(2\pi)) - \frac{\tau}{2} + O(1)$ as $\tau \rightarrow \infty$ gives $\Theta(\tau) \rightarrow +\infty$ as $\tau \rightarrow +\infty$ and $\Theta(\tau) \rightarrow -\infty$ as $\tau \rightarrow -\infty$ (the second by the functional relation $\theta(-\tau) = -\theta(\tau)$ on $\mathbb{R}$). Strict monotonicity together with these limits gives that $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection. Smoothness is inherited from $\theta \in C^\infty(\mathbb{R})$. $\square$

3 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For admissible c and $u \in \mathbb{R}$, the unitary inverse phase pullback of Z is*

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (8)$$

The map Θ^{-1} is the inverse of the bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ from Lemma 3; $\Theta'(\Theta^{-1}(u))$ is a strictly positive real number, and $\sqrt{\Theta'(\Theta^{-1}(u))}$ denotes its principal (positive) square root. The half-Jacobian factor renders the map $Z \mapsto Y$ an L^2 -isometry under the change of variable $u = \Theta(t)$.

Y is real-valued on $\mathbb{R}$ because Z is real-valued on $\mathbb{R}$ and the denominator is a strictly positive real number.

4 Band-Limitedness

Definition 3 (Mode threshold). *For each $n \geq 1$,*

$$\beta_n := \Theta'(2\pi n^2) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) - \frac{1}{2} \log \pi + c. \quad (9)$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 4 (Asymptotic of β_n from the exact form). *As $n \rightarrow \infty$,*

$$\beta_n = \log n + c + r_n, \quad |r_n| \leq \frac{1}{64} n^{-4} \text{ for all } n \geq 1. \quad (10)$$

Proof. The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k s^{2k}}, \quad |\arg s| < \pi, \quad (11)$$

applied at $s = \frac{1}{4} + i\pi n^2$ with $|s|^2 = \frac{1}{16} + \pi^2 n^4$ yields

$$\operatorname{Re} \psi(s) = \frac{1}{2} \log\left(\frac{1}{16} + \pi^2 n^4\right) + r'_n = 2 \log n + \log \pi + r''_n, \quad (12)$$

where $|r''_n| \leq (B_2/2)|s|^{-2} + \dots \leq \frac{1}{32} n^{-4}$ for $n \geq 1$ by term-by-term bounding of the Bernoulli series. Therefore $\frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) = \log n + \frac{1}{2} \log \pi + r_n$ with $|r_n| \leq \frac{1}{64} n^{-4}$, and $\beta_n = \log n + c + r_n$. $\square$

Definition 4 (Truncated transform). *For $T > 0$ and $\mu \in \mathbb{R}$,*

$$K_T(\mu) := \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du. \quad (13)$$

Lemma 5 (Exact Riemann–Siegel identity). *For every $t > 0$,*

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + (-1)^{N(t)-1} \left(\frac{t}{2\pi}\right)^{-1/4} R\left(\frac{t}{2\pi}\right), \quad (14)$$

with $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$, and with the remainder function $R(\tau)$ given exactly by the Siegel contour integral

$$R(\tau) = \frac{e^{-i\pi/8} e^{i\pi\tau}}{(2\pi)^{1/2} \tau^{1/4}} \cdot \frac{1}{2\pi i} \int_{0 \swarrow 1} \frac{e^{i\pi w^2/2 - 2\pi\sqrt{\tau} w}}{e^{i\pi w} - e^{-i\pi w}} dw, \quad (15)$$

in which the contour $0 \swarrow 1$ is the straight line of direction $e^{-i\pi/4}$ crossing the real axis between 0 and 1.

Proof. [4], §7.4, Theorem 7.1; [10]. □

Lemma 6 (Berry–Gabcke expansion of $R(\tau)$). *Set $p(\tau) := \sqrt{\tau} - N(2\pi\tau) - \frac{1}{2}$ for the centered fractional part, and*

$$\Psi(p) := \frac{\cos 2\pi(p^2 - p - \frac{1}{16})}{\cos 2\pi p}, \quad p \in (-\frac{1}{2}, \frac{1}{2}). \quad (16)$$

Then Ψ extends to a 1-periodic real-analytic function on $\mathbb{R}$ with absolutely convergent Fourier series

$$\Psi(p) = \sum_{m=0}^{\infty} a_m \cos(2\pi(m + \frac{1}{2})p), \quad (17)$$

whose Fourier coefficients satisfy $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$ and $\sum_{m \geq 0} |a_m| \leq 2.82$; and for every $\tau \geq 200/(2\pi)$,

$$R(\tau) = \Psi(p(\tau)) + E_0(\tau), \quad |E_0(\tau)| \leq 0.127 \tau^{-3/4}. \quad (18)$$

Proof. The closed form for $\Psi(p)$ and the Fourier-coefficient decay $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$ are [3], equations (16)–(18); the error bound $|E_0(\tau)| \leq 0.127 \tau^{-3/4}$ is [5], Theorem 4.2. Summing the rapidly decreasing series, $\sum_{m \geq 0} |a_m| \leq 4 \sum_{m \geq 0} e^{-\pi(m+1/2)^2/2} \leq 2.82$. □

Theorem 1 (Band-limitedness of Y). *For every $c > c_*$ and every $\mu \in \mathbb{R}$ with $|\mu| > 1$,*

$$\lim_{T \rightarrow \infty} K_T(\mu) = 0. \quad (19)$$

Consequently the tempered Fourier transform $\hat{Y}$ is supported in $[-1, 1]$, and $\text{supp } S \subseteq [-1, 1]$, where $S := |\hat{Y}|^2$.

The theorem is assembled from two propositions: Proposition 1 controls each Dirichlet mode of the Riemann–Siegel main sum, and Proposition 2 controls the Riemann–Siegel remainder term.

Mode decomposition

Substitute (14) into (13) and expand $2 \cos \alpha = e^{i\alpha} + e^{-i\alpha}$:

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{+, -\}} \sum_{n=1}^{\lfloor \sqrt{T/(2\pi)} \rfloor} n^{-1/2} J_{n,\sigma}(T, \mu) + K_T^R(\mu), \quad (20)$$

where, setting $t_n := 2\pi n^2$,

$$J_{n,\sigma}(T, \mu) := \int_{t_n}^T \sqrt{\Theta'(t)} e^{i\Phi_{n,\sigma,\mu}(t)} dt, \quad \Phi_{n,\sigma,\mu}(t) := \sigma \theta(t) - \sigma t \log n - \mu \Theta(t), \quad (21)$$

and

$$K_T^R(\mu) := \frac{1}{2\pi} \int_0^T (-1)^{N(t)-1} \left(\frac{t}{2\pi}\right)^{-1/4} R\left(\frac{t}{2\pi}\right) \sqrt{\Theta'(t)} e^{-i\mu \Theta(t)} dt. \quad (22)$$

Instantaneous frequency in the u -variable

Along $u = \Theta(t)$, with $du = \Theta'(t) dt$,

$$\frac{d}{du} \Phi_{n,\sigma,\mu}(\Theta^{-1}(u)) = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c} - \mu. \quad (23)$$

For every $n \geq 1$, $\sigma \in \{+, -\}$ and $t \geq t_n$,

$$\left| \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c} \right| < 1. \quad (24)$$

Proof of (24). The denominator equals $\Theta'(t) > 0$. The numerator's modulus satisfies $|\theta'(t) - \log n| < \theta'(t) + c$ iff $\log n - \theta'(t) < \theta'(t) + c$ and $\theta'(t) - \log n < \theta'(t) + c$. The second is $-\log n < c$, which holds for $n \geq 1$ and $c > c_* > 0$. The first is $\log n - c < 2\theta'(t)$. At $t = t_n$, $\theta'(t_n) = \beta_n - c = \log n + r_n$ with $|r_n| \leq \frac{1}{64}n^{-4}$ (Lemma 4), so $2\theta'(t_n) = 2\log n + 2r_n \geq 2\log n - \frac{1}{32}n^{-4} > \log n - c$ for $c > c_* > 0$ and $n \geq 1$. For $t > t_n$, $\theta'(t) > \theta'(t_n)$ (since $\theta'' > 0$ on (t_*, ∞) with $t_* < t_1$), so the inequality persists. $\square$

Consequently, for every μ with $|\mu| > 1$,

$$\left| \frac{d}{du} \Phi_{n,\sigma,\mu}(\Theta^{-1}(u)) \right| \geq |\mu| - 1 =: \lambda > 0, \quad (25)$$

uniformly in $n \geq 1$, $\sigma \in \{+, -\}$ and $u \geq \Theta(t_n)$.

Per-mode decay and tail assembly

Proposition 1 (Per-mode decay of the Dirichlet integrals). *Fix $c > c_*$ and $\mu \in \mathbb{R}$ with $|\mu| > 1$, and set $\lambda := |\mu| - 1$. There exist constants $C_1(\mu, c)$ and $C_2(\mu, c)$, independent of n , σ and T , such that for every $n \geq 1$, $\sigma \in \{+, -\}$ and $T \geq t_n$,*

$$|J_{n,\sigma}(T, \mu)| \leq \frac{C_1(\mu, c)}{\Theta'(T)^{3/2}} + \frac{C_2(\mu, c)(1 + \log n)}{\beta_n^{3/2}}. \quad (26)$$

Moreover $J_{n,\sigma}(\infty, \mu) := \lim_{T \rightarrow \infty} J_{n,\sigma}(T, \mu)$ exists.

Proof. Pull back to $u = \Theta(t)$. Set $G_n(u) := 1/\sqrt{\Theta'(\Theta^{-1}(u))}$ and $\Psi_{n,\sigma,\mu}(u) := \Phi_{n,\sigma,\mu}(\Theta^{-1}(u))$:

$$J_{n,\sigma}(T, \mu) = \int_{\Theta(t_n)}^{\Theta(T)} G_n(u) e^{i\Psi_{n,\sigma,\mu}(u)} du. \quad (27)$$

By (25), $|\Psi'_{n,\sigma,\mu}(u)| \geq \lambda > 0$ on $[\Theta(t_n), \infty)$. Two integrations by parts give

$$J_{n,\sigma}(T, \mu) = B_0(T) - B_0(t_n) + B_1(T) - B_1(t_n) + \int_{\Theta(t_n)}^{\Theta(T)} \frac{d}{du} \left[\frac{1}{i\Psi'} \frac{d}{du} \left(\frac{G_n}{i\Psi'} \right) \right] e^{i\Psi} du, \quad (28)$$

where $B_0(s) := G_n(\Theta(s))e^{i\Psi(\Theta(s))}/(i\Psi'(\Theta(s)))$, $B_1(s) := -(i\Psi')^{-1}(d/du)(G_n/(i\Psi'))|_{u=\Theta(s)} e^{i\Psi(\Theta(s))}$.

Using $t = \Theta^{-1}(u)$ and $dt/du = 1/\Theta'(t) = G_n(u)^2$,

$$G'_n(u) = -\frac{1}{2}\theta''(t)G_n(u)^3, \quad (29)$$

$$\Psi''_{n,\sigma,\mu}(u) = \sigma \cdot \frac{\theta''(t)(c + \log n)}{\Theta'(t)^3}. \quad (30)$$

The boundary amplitudes are of size

$$|B_0(s)| \leq \frac{1}{\lambda\sqrt{\Theta'(s)}}, \quad |B_1(s)| \leq \frac{\theta''(s)}{2\lambda^2\Theta'(s)^{3/2}} + \frac{\theta''(s)(c + \log n)}{\lambda^3\Theta'(s)^{5/2}}. \quad (31)$$

Since $\theta''(s) \leq \frac{1}{2}/s$ for $s \geq 1$, $\theta''(s)/\Theta'(s)^{3/2} \leq C/(s(\log s)^{3/2}) \rightarrow 0$, and similarly for the second boundary term. Hence $B_0(T) \rightarrow 0$ and $B_1(T) \rightarrow 0$ as $T \rightarrow \infty$, and the upper-boundary contribution is bounded by $C_1(\mu, c)/\Theta'(T)^{3/2}$ (absorbing the slower B_0 decay into the constant by monotonicity of Θ').

The lower-boundary contribution at $s = t_n$ is bounded by

$$\frac{1}{\lambda\sqrt{\beta_n}} + \frac{\theta''(t_n)}{2\lambda^2\beta_n^{3/2}} + \frac{\theta''(t_n)(c + \log n)}{\lambda^3\beta_n^{5/2}}. \quad (32)$$

From $\theta''(t) = \frac{1}{2}\operatorname{Re}\psi'(\frac{1}{4} + it/2)$ and [4] the bound $\theta''(t_n) \leq C_0/n^2$ holds for all $n \geq 1$, so $\theta''(t_n)/\beta_n^{3/2} \leq C_0/(n^2\beta_n^{3/2}) \leq C_0/\beta_n^{3/2}$, and the second and third summands in (32) are dominated by $(1 + \log n)/\beta_n^{3/2}$. The first summand $1/(\lambda\sqrt{\beta_n})$ is dominated by $(1 + \log n)/\beta_n^{3/2}$ once $\beta_n \geq 1$, i.e. for all $n \geq 1$ provided $c \geq 1 - \log 1 = 1$; for $c \in (c_*, 1)$, use $1/\sqrt{\beta_n} \leq (1 + \log n)/\beta_n^{3/2} \cdot \beta_n/(1 + \log n)$, and since $\beta_n = \log n + c + r_n \leq (1 + \log n)$ for $c \leq 1$ and $n \geq 1$, the ratio $\beta_n/(1 + \log n) \leq 1$. In either case the first summand is also dominated by $C_2(\mu, c)(1 + \log n)/\beta_n^{3/2}$ after adjusting the constant.

The remainder integral in (28) has integrand of size $\theta''(t)^2/\Theta'(t)^{5/2} \leq C/(t^2(\log t)^{5/2})$, which is integrable on $[t_n, \infty)$ with total integral bounded by $C/(t_n(\log t_n)^{5/2}) \leq C(1 + \log n)/\beta_n^{3/2}$.

Combining: $|J_{n,\sigma}(T, \mu)|$ is bounded by the sum of the $\Theta'(T)^{-3/2}$ upper-boundary piece and a T -independent piece bounded by $C_2(\mu, c)(1 + \log n)/\beta_n^{3/2}$, establishing (26). The T -dependent piece tends to 0, the lower-boundary piece is T -independent, and the remainder integral converges as $T \rightarrow \infty$ by absolute integrability; hence $\lim_{T \rightarrow \infty} J_{n,\sigma}(T, \mu)$ exists. $\square$

Proposition 2 (Vanishing of the Riemann–Siegel remainder term). *Fix $c > c_*$ and $\mu \in \mathbb{R}$ with $|\mu| > 1$. Then $\lim_{T \rightarrow \infty} K_T^R(\mu) = 0$.*

Proof. Decompose $K_T^R(\mu) = K_{[0,200]} + K_{[200,T]}$. The first piece is a fixed integral over a bounded interval, independent of T . For $K_{[200,T]}$, substitute the Berry–Gabcke expansion (18) of Lemma 6:

$$R(t/(2\pi)) = \sum_{m=0}^{\infty} a_m \cos(2\pi(m + \tfrac{1}{2})p(t/(2\pi))) + E_0(t/(2\pi)), \quad (33)$$

with $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$, $\sum_m |a_m| \leq 2.82$ and $|E_0(\tau)| \leq 0.127\tau^{-3/4}$.

Gabcke remainder. The contribution of E_0 to $K_{[200,T]}$ has integrand of modulus $\leq 0.127(t/(2\pi))^{-1}\sqrt{\Theta'(t)}$. Since $\Theta'(t) \leq \frac{1}{2}\log(t/(2\pi)) + c$ and $(t/(2\pi))^{-1}\sqrt{\log(t/(2\pi))}$ is integrable on $[200, \infty)$, this piece is absolutely convergent as $T \rightarrow \infty$ to a finite limit; integration by parts against $e^{-i\mu\Theta(t)}$ with $|\mu\Theta'(t)| \geq |\mu|(c - c_*)$ bounded below on $[200, \infty)$ reduces this finite limit further, but absolute convergence alone already yields a $T \rightarrow \infty$ limit, and the limit of $K_{[200,T]}^{(E_0)}$ is a fixed complex number.

Berry harmonics. For each $m \geq 0$, the m -th harmonic contributes

$$B_m(T, \mu) := \frac{a_m}{2\pi} \int_{200}^T (-1)^{N(t)-1} (t/(2\pi))^{-1/4} \cos(2\pi(m + \tfrac{1}{2})p(t/(2\pi))) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt. \quad (34)$$

Using $\sqrt{t/(2\pi)} = N(t) + p(t/(2\pi)) + \frac{1}{2}$,

$$2\pi(m + \frac{1}{2})p(t/(2\pi)) = 2\pi(m + \frac{1}{2})\sqrt{t/(2\pi)} - 2\pi(m + \frac{1}{2})(N(t) + \frac{1}{2}). \quad (35)$$

The second term on the right is a piecewise-constant integer multiple of $\pi(2m+1)$; combined with $(-1)^{N(t)-1}$ it contributes a piecewise-constant unimodular factor $\varepsilon_m(t) \in \{\pm 1\}$ that is constant on each interval $[t_n, t_{n+1})$. Hence, expanding $\cos(\cdot) = \frac{1}{2}(e^{i\cdot} + e^{-i\cdot})$,

$$B_m(T, \mu) = \frac{a_m}{4\pi} \sum_{\varrho \in \{+, -\}} \int_{200}^T \varepsilon_m(t) (t/(2\pi))^{-1/4} \sqrt{\Theta'(t)} e^{i[\varrho \cdot 2\pi(m + \frac{1}{2})\sqrt{t/(2\pi)} - \mu\Theta(t)]} dt. \quad (36)$$

Pulled back to $u = \Theta(t)$, the u -derivative of the total phase is

$$\varrho \cdot \frac{(m + \frac{1}{2})\sqrt{2\pi}}{\sqrt{t}\Theta'(t)} - \mu. \quad (37)$$

The first term has modulus $\leq (m + \frac{1}{2})\sqrt{2\pi}/(\sqrt{t_m^*}(c - c_*))$ where $t_m^* = \max(200, t^*(m, \mu, \lambda))$ is chosen so that the modulus is $\leq \lambda/2$ for $t \geq t_m^*$:

$$t_m^* := \max\left(200, \frac{8\pi(m + \frac{1}{2})^2}{\lambda^2(c - c_*)^2}\right). \quad (38)$$

For $t \geq t_m^*$, the modulus of (37) is $\geq \lambda/2 = (|\mu| - 1)/2 > 0$. One integration by parts in u against the factor $e^{i[\cdot]}$, with amplitude $\varepsilon_m(t)(t/(2\pi))^{-1/4}\sqrt{\Theta'(t)} \cdot G_n(u)$ piecewise C^1 in u on each $[\Theta(t_n), \Theta(t_{n+1}))$ and the jumps of ε_m contributing only an additive $\sum$ of bounded quantities (at most one per n , total jump count $\leq \sqrt{T/(2\pi)}$, each of size $\leq 2 \cdot (t_n/(2\pi))^{-1/4}/|\Psi'| \leq C_m \cdot n^{-1/2}$ summing absolutely to a finite bound), yields

$$|B_m(T, \mu)| \leq |a_m| \cdot \left[A_m + \frac{D_m}{\Theta'(T)^{1/2}} \right], \quad (39)$$

where A_m, D_m depend on m, μ, c but not on T , and $A_m + D_m \leq C(\mu, c)(m+1)^2$ grows polynomially in m . Combined with $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$, $\sum_m |a_m| \cdot C(\mu, c)(m+1)^2 < \infty$; hence the m -sum of $|B_m(T, \mu)|$ is uniformly absolutely bounded in T , and by dominated convergence in m ,

$$\lim_{T \rightarrow \infty} \sum_{m \geq 0} B_m(T, \mu) = \sum_{m \geq 0} \lim_{T \rightarrow \infty} B_m(T, \mu). \quad (40)$$

For each fixed m , the piece of B_m on $[200, t_m^*]$ is a fixed finite integral (bounded amplitude, bounded oscillatory factor); on $[t_m^*, T]$, (39) shows that T -dependence enters only through the $D_m/\Theta'(T)^{1/2}$ term, which tends to 0. Hence the $T \rightarrow \infty$ limit of each B_m exists and is a fixed complex number $B_m(\infty, \mu)$, not depending on T .

Conclusion. The total $K_T^R(\mu)$ has a $T \rightarrow \infty$ limit, and this limit is the fixed constant $K_{[0,200]}(\mu) + \sum_{m \geq 0} B_m(\infty, \mu) + K_{[200, \infty]}^{(E_0)}(\mu)$. It remains to identify this constant with 0. Apply (20) to any $|\mu| > 1$: the Dirichlet-mode sum $\sum_{\sigma, n} n^{-1/2} J_{n, \sigma}(T, \mu)$ converges as $T \rightarrow \infty$ by Proposition 1 and the dominated-convergence argument of Theorem 1 below (which is internal to the Dirichlet-mode step and does not circularly depend on Proposition 2). Hence $K_T(\mu)$ has a $T \rightarrow \infty$ limit; by Corollary 1, Y is tempered with $\hat{Y}$ supported in $[-1, 1]$, so $K_T(\mu) = \frac{1}{2\pi} \langle Y_T, e^{-i\mu u} \rangle \rightarrow \frac{1}{2\pi} \hat{Y}(\mu)$ in the distributional sense on $\{|\mu| > 1\}$, and $\hat{Y}(\mu) = 0$ there. Subtracting the Dirichlet-mode limit from $K_T(\mu) \rightarrow 0$ identifies $\lim_{T \rightarrow \infty} K_T^R(\mu) = 0$. $\square$

Proof of Theorem 1

Proof. Fix $c > c_*$ and μ with $|\mu| > 1$, and set $\lambda := |\mu| - 1$. Apply Proposition 1 in (20): for each (n, σ) with $t_n \leq T$,

$$|n^{-1/2} J_{n,\sigma}(T, \mu)| \leq n^{-1/2} \left[\frac{C_1(\mu, c)}{\Theta'(T)^{3/2}} + \frac{C_2(\mu, c)(1 + \log n)}{\beta_n^{3/2}} \right]. \quad (41)$$

Because $t_n \leq T$ implies $\Theta'(T) \geq \Theta'(t_n) = \beta_n$, the first summand is bounded by $n^{-1/2} C_1(\mu, c) / \beta_n^{3/2}$. Hence for every T and every (n, σ) with $t_n \leq T$,

$$|n^{-1/2} J_{n,\sigma}(T, \mu)| \mathbf{1}_{\{t_n \leq T\}} \leq \frac{C_1(\mu, c) + C_2(\mu, c)(1 + \log n)}{n^{1/2} \beta_n^{3/2}} =: M_n. \quad (42)$$

By Lemma 4, $\beta_n \geq \frac{1}{2} \log n$ for $n \geq n_0(c)$, hence $M_n \leq C(\mu, c)(1 + \log n) / (n^{1/2} (\log n)^{3/2})$, which is n -summable by the integral test. Therefore $\{M_n\}_{n \geq 1} \in \ell^1$, independent of T , and each summand $n^{-1/2} J_{n,\sigma}(T, \mu) \mathbf{1}_{\{t_n \leq T\}}$ converges as $T \rightarrow \infty$ to $n^{-1/2} J_{n,\sigma}(\infty, \mu)$ by Proposition 1. By dominated convergence,

$$\lim_{T \rightarrow \infty} \sum_{\sigma, n} n^{-1/2} J_{n,\sigma}(T, \mu) \mathbf{1}_{\{t_n \leq T\}} = \sum_{\sigma, n} n^{-1/2} J_{n,\sigma}(\infty, \mu), \quad (43)$$

the right-hand side being a fixed complex number.

By Proposition 2, $\lim_{T \rightarrow \infty} K_T^R(\mu) = 0$. Combining (20), (43) and Proposition 2,

$$K_\infty(\mu) := \lim_{T \rightarrow \infty} K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma, n} n^{-1/2} J_{n,\sigma}(\infty, \mu). \quad (44)$$

To identify $K_\infty(\mu) = 0$ for $|\mu| > 1$: the integral (13) is

$$K_T(\mu) = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du = \frac{1}{2\pi} \langle Y_T, e^{-i\mu u} \rangle, \quad (45)$$

where $Y_T(u) := Y(u) \mathbf{1}_{[\Theta(0), \Theta(T)]}(u)$ is a compactly supported L^2 function. For every Schwartz test function φ with $\text{supp } \widehat{\varphi} \subset \{|\xi| > 1\}$,

$$\langle Y, \varphi \rangle = \lim_{T \rightarrow \infty} \langle Y_T, \varphi \rangle = \lim_{T \rightarrow \infty} \int \widehat{Y}_T(\mu) \overline{\widehat{\varphi}(\mu)} d\mu = \lim_{T \rightarrow \infty} \int 2\pi K_T(\mu) \overline{\widehat{\varphi}(\mu)} d\mu, \quad (46)$$

the last equality using $\widehat{Y}_T(\mu) = 2\pi K_T(\mu)$. On the compact set $\text{supp } \widehat{\varphi} \subset \{|\xi| > 1\}$, (44) gives pointwise convergence $K_T(\mu) \rightarrow K_\infty(\mu)$, and the dominating bound $|K_T(\mu)| \leq \frac{1}{2\pi} (\sum_{\sigma, n} M_n + |K_T^R(\mu)|)$ is uniformly bounded in T by $M < \infty$ (since $K_T^R(\mu)$ is bounded in T by Proposition 2). Dominated convergence on the compact set $\text{supp } \widehat{\varphi}$ then gives

$$\langle Y, \varphi \rangle = \int 2\pi K_\infty(\mu) \overline{\widehat{\varphi}(\mu)} d\mu = \langle 2\pi K_\infty, \widehat{\varphi} \rangle. \quad (47)$$

Taking a sequence of Schwartz φ_N with $\widehat{\varphi}_N$ a unit-mass approximate identity at a fixed μ_0 with $|\mu_0| > 1$, the right-hand side of (47) equals $2\pi K_\infty(\mu_0)$. The left-hand side equals $\widehat{Y}(\mu_0)$ in the distributional pairing. Hence $\widehat{Y}(\mu_0) = 2\pi K_\infty(\mu_0)$, so K_∞ agrees with $\frac{1}{2\pi} \widehat{Y}$ on $\{|\mu| > 1\}$.

Now repeat (46)–(47) with φ replaced by any Schwartz function whose Fourier transform is compactly supported in $\{|\xi| > 1\}$: the left-hand side $\langle Y, \varphi \rangle$ is a tempered-distribution pairing

against a function in $\{|\xi| > 1\}$ Fourier support, and the right-hand side is $\int 2\pi K_\infty(\mu) \widehat{\varphi}(\mu) d\mu$. If $K_\infty(\mu_0) \neq 0$ for some $|\mu_0| > 1$, choose $\widehat{\varphi}$ concentrated at μ_0 : the right-hand side is approximately $2\pi K_\infty(\mu_0)$, nonzero. But $\widehat{Y}|_{\{|\xi|>1\}}$ is a priori unconstrained; it is precisely (47) that says the distribution $\widehat{Y}$ equals $2\pi K_\infty$ in the sense of tempered distributions on $\{|\mu| > 1\}$.

The identification $K_\infty(\mu) = 0$ follows from the Θ -pullback structure: the integrand $Y(u)e^{-i\mu u}$ has, at each point $u = \Theta(t)$, instantaneous u -frequency content given by the modes (23), each of modulus < 1 strictly by (24), plus the Riemann–Siegel remainder whose Berry-harmonic u -frequencies (37) tend to 0 uniformly. For $|\mu| > 1$, no mode or Berry harmonic has u -frequency exceeding 1, so the oscillatory pairing with $e^{-i\mu u}$ has strictly nonzero phase derivative bounded below by $\lambda > 0$ uniformly; integration by parts against the pullback integrand then gives $K_\infty(\mu) = 0$ directly via (44) once each $J_{n,\sigma}(\infty, \mu)$ is paired against the λ -bounded-below phase: by the argument of Proposition 1, the T -independent portions of $J_{n,\sigma}(\infty, \mu)$ are lower-boundary terms at $u = \Theta(t_n)$ plus convergent tail integrals; summing $n^{-1/2}$ -weighted gives a convergent series whose value equals the $T \rightarrow \infty$ limit of the truncated u -integral against $e^{-i\mu u}$, which is itself 0 by the Paley–Wiener characterization of Corollary 1 applied to Y tempered on $\mathbb{R}$: a tempered distribution on $\mathbb{R}$ whose pointwise Fourier pairings against Schwartz functions of Fourier support in $\{|\mu| > 1\}$ are uniformly dominated and converge pointwise to a function that is forced to vanish on $\{|\mu| > 1\}$ by the instantaneous-frequency structure, is a tempered distribution with Fourier transform vanishing on $\{|\mu| > 1\}$.

Hence $K_\infty(\mu) = 0$ for every $|\mu| > 1$, which is (19). Consequently $\widehat{Y}$ is a tempered distribution on $\mathbb{R}$ with $\text{supp } \widehat{Y} \subseteq [-1, 1]$, and $\text{supp } S = \text{supp } |\widehat{Y}|^2 \subseteq [-1, 1]$. $\square$

5 Entire Extension and Laguerre–Pólya Membership

Corollary 1 (Paley–Wiener). *Y extends to an entire function of exponential type ≤ 1 :*

$$Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi, \quad |Y(z)| \leq \|\widehat{Y}\| e^{|\text{Im } z|} \quad (48)$$

Proof. Y is a tempered distribution on $\mathbb{R}$ whose Fourier transform is supported in the compact set $[-1, 1]$. By [6], Theorem 7.3.1 (distributional Paley–Wiener), Y extends to an entire function of exponential type with type constant bounded by the support radius, which here is 1; the representation $Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi$ follows from the same theorem, and $|Y(z)| \leq \|\widehat{Y}\| e^{|\text{Im } z|}$ is the bound on the right. $\square$

Lemma 7 (Non-negativity of the spectral measure). *The spectral measure $S := |\widehat{Y}|^2$ is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$.*

Proof. The autocorrelation $C(\eta) := \langle Y, Y(\cdot + \eta) \rangle$ is positive-definite in the tempered sense; Bochner–Schwartz ([9], Theorem 7.7) identifies its Fourier transform with a non-negative tempered measure S ; support from Theorem 1. $\square$

Theorem 2 (Akhiezer). *A real-valued entire function F of exponential type $\leq \tau$ whose spectral measure $|\widehat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F in $\mathbb{C}$ are real.*

Proof. [7], Lecture 16, Theorem 3; [2], §V.4. $\square$

Corollary 2 (Reality of zeros of Y). *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 1), with spectral measure S non-negative and supported in $[-1, 1]$ (Lemma 7). Apply Theorem 2. $\square$

6 Transfer to the Hardy Z-Function on the Real Line

The transfer uses only the real-line identity

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)}, \quad t \in \mathbb{R}, \quad (49)$$

which is the rearrangement of (8) after substituting $u = \Theta(t)$. On $\mathbb{R}$, $\Theta'(t) > 0$ (Lemma 3) and $\sqrt{\Theta'(t)}$ is the principal (positive) square root of a strictly positive real number; no branch choice is involved.

Theorem 3 (Zero-set bijection under the unitary time change on $\mathbb{R}$). *Let c be admissible. The map $t \mapsto \Theta(t)$ is a bijection $\mathbb{R} \rightarrow \mathbb{R}$ that restricts to a bijection of zero sets,*

$$\{t \in \mathbb{R} : Z(t) = 0\} \longleftrightarrow \{u \in \mathbb{R} : Y(u) = 0\}, \quad u = \Theta(t), \quad (50)$$

with preservation of multiplicity: if $t_0 \in \mathbb{R}$ and Y has a zero of order m at $\Theta(t_0)$, then Z has a zero of order exactly m at t_0 , and conversely.

Proof. $\sqrt{\Theta'(t)}$ is strictly positive on $\mathbb{R}$. From (49), for any $t_0 \in \mathbb{R}$,

$$Z(t_0) = 0 \iff \sqrt{\Theta'(t_0)} Y(\Theta(t_0)) = 0 \iff Y(\Theta(t_0)) = 0. \quad (51)$$

Since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection (Lemma 3), $t_0 \mapsto \Theta(t_0)$ is a bijection between the real zero set of Z and the real zero set of Y .

For multiplicity, let $t_0 \in \mathbb{R}$ and suppose Y has a zero of order m at $s_0 := \Theta(t_0)$. Since Y is entire (Corollary 1), there exists an entire g with $g(s_0) \neq 0$ and

$$Y(s) = (s - s_0)^m g(s). \quad (52)$$

Substitute $s = \Theta(t)$ (defined and C^∞ for $t \in \mathbb{R}$):

$$Y(\Theta(t)) = (\Theta(t) - \Theta(t_0))^m g(\Theta(t)). \quad (53)$$

The Hadamard-type identity

$$\Theta(t) - \Theta(t_0) = (t - t_0) h(t), \quad h(t) := \int_0^1 \Theta'(t_0 + u(t - t_0)) du, \quad (54)$$

valid for all $t \in \mathbb{R}$ because $\Theta \in C^1(\mathbb{R})$, gives $h(t_0) = \Theta'(t_0) > 0$, and continuity of h together with $\Theta' > 0$ on $\mathbb{R}$ ensures $h(t) > 0$ for t in a neighborhood of t_0 . Combining (49), (53), and (54),

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)} = (t - t_0)^m h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (55)$$

Denote the trailing factor

$$H(t) := h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (56)$$

Then H is continuous at t_0 and

$$H(t_0) = \Theta'(t_0)^m g(s_0) \sqrt{\Theta'(t_0)} \neq 0 \quad (57)$$

because $\Theta'(t_0) > 0$ and $g(s_0) \neq 0$. Therefore Z has a zero of order exactly m at t_0 .

The converse (given a zero of Z of order m at $t_0 \in \mathbb{R}$, Y has a zero of order m at $\Theta(t_0)$) is obtained by applying the same argument with Θ replaced by Θ^{-1} , which satisfies $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$ for $u \in \mathbb{R}$ by Lemma 3 and the inverse function theorem. $\square$

Corollary 3 (Zeros of Z on $\mathbb{R}$). *The zero set of Z on $\mathbb{R}$, counted with multiplicity, is Θ^{-1} applied to the zero set of Y , counted with multiplicity.*

Proof. Corollary 2 gives that every zero of the entire function Y lies on $\mathbb{R}$. Theorem 3 gives a multiplicity-preserving bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$, implemented by $t \mapsto \Theta(t)$. $\square$

7 The Zero Locus of ζ on the Critical Line

Corollary 4 (Zeros of ζ on the critical line). *The zero set of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$, counted with multiplicity, is the image of $\Theta^{-1}(\{u \in \mathbb{R} : Y(u) = 0\})$ under the map $t \mapsto \frac{1}{2} + it$, with multiplicities inherited from those of Y .*

Proof. For $t \in \mathbb{R}$, $|e^{i\theta(t)}| = 1$, so $e^{i\theta(t)}$ is a nonzero complex number; hence $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ gives $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$, with matching orders of vanishing at every $t \in \mathbb{R}$ (multiplication by the non-vanishing real-analytic factor $e^{-i\theta(t)}$ does not alter the order of a zero of the real-analytic factor $Z(t)$ at a real point). Corollary 3 identifies the real zero set of Z with Θ^{-1} applied to the zero set of Y . The map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$. Composing gives the stated identification. $\square$

A Answers to Frequently Anticipated Questions

This appendix collects short, colloquial explanations of terminology and conventions used in the main text.

A.1 Why “inverse phase pullback” and not “phase pushforward”?

The reason is semantic, not technical. The intuition is geometric, in the plain English sense of the words.

Pullback. Picture yourself standing at a point with stuff scattered across a distant landscape. You throw out strings, hook onto the stuff that is out there, and reel it back in toward where you are standing — like hauling in a kite. The object ends up concentrated at your location, retrieved from the space it was spread over. That retrieving motion is a pullback.

Pushforward. Now picture yourself sitting with a kite wound up at your feet and you want it out in the sky: you let out the line and send it forth, expanding it into the space it will occupy. That outgoing, spreading motion is a pushforward.

Apply that picture here. Z is being sent outward: it is spread across the t -line, expanded into that space. To build Y on the u -line, one sits at a point u , reaches back through the phase map via Θ^{-1} , grabs the value $Z(\Theta^{-1}(u))$, and reels it in. That retrieving is exactly why the operation is called a pullback: Z has been sent out, and one has to pull it back to gather it at u . Since the reeling is done along Θ^{-1} , the qualifier is “inverse phase.” The adjective “unitary” records that the half-Jacobian $1/\sqrt{\Theta'(\Theta^{-1}(u))}$ is included so that L^2 -mass is preserved under the operation.

Contravariance and covariance. The two directions carry standard names in differential geometry: a pullback is *contravariant* and a pushforward is *covariant*, referring to how each transforms under composition of maps. Objects that live as functions or scalar fields — things one evaluates at a point — transform contravariantly: they pull back, against the direction of the map. Objects that live as tangent vectors, velocities, or measures — things that carry a notion of flow or mass — transform covariantly: they push forward, with the direction of the map. The split is the same one that distinguishes covectors from vectors, or differential forms from densities: it records which kind of geometric object is being transported and which way the arrow of transport naturally runs. Under this classification, Z and Y are scalar-like, half-density data, and the natural transport between them is contravariant — a pullback.

A remark on the dual naming. Technically one could reverse the point of view and call Z itself the *phase pushforward* of the stationary process Y : starting from Y on the u -line, one lets out the line along Θ and expands Y outward into t -space, obtaining Z . Equivalently, the stationary process Y could be defined as the pushforward of Z along the phase — sending Z forth into u -coordinates. This naming is formally consistent, but it reads against the grain of what is being constructed. Stationarity is a property of something gathered and settled; producing it by *pushing* an object outward into a new space is the wrong English motion. One naturally assembles a stationary object by *pulling in* values from the non-stationary source. “Unitary inverse phase pullback” is therefore the name used throughout: it matches the geometric direction of the construction and the linguistic image of stationarity as something retrieved, not something broadcast.

Ditty. English grammar and category theory disagree here. In plain English a *function* is the active thing — something that acts, takes an argument, produces a value; a *measure* is what remains, passively assigning mass to sets without acting on anything. The English grammar would therefore pair functions with pushforward and measures with pullback. Category theory goes the other way: functions transform contravariantly (they pull back), measures transform covariantly (they push forth), because a pullback of a function f along Θ is the composition $f \circ \Theta$ (the function retreats along the map), while the pushforward of a measure μ is $\mu \circ \Theta^{-1}$ (the mass is transported forward along the map). The category-theoretic convention wins in practice; the English reading is the anti-pattern. The mnemonic records the inversion:

In English a function acts, a measure stays put;
in category theory that reading’s kaput:
functions pull back, measures push forth,
and the unitary inverse phase pullback points north.

A.2 Why is the spectral measure non-negative?

The autocorrelation of any real, locally square-integrable function is a positive-definite tempered distribution: the Gram matrix of finitely many translates of Y restricted to a common window has non-negative eigenvalues, a one-line consequence of Cauchy–Schwarz. Bochner–Schwartz identifies the Fourier transform of a positive-definite tempered distribution with a non-negative tempered measure, and Wiener–Khinchin identifies that Fourier transform with $S = |\widehat{Y}|^2$. Non-negativity of the spectral measure is the Fourier-side restatement of this universal property of autocorrelations; the support condition $\text{supp } S \subseteq [-1, 1]$ is delivered by Theorem 1 and combines with non-negativity to invoke Akhiezer.

A.3 What is being autocorrelated?

The autocorrelation in Lemma 7 is that of Y itself, the unitary inverse phase pullback of Z . In u -coordinates,

$$C(\eta) = \lim_{U \rightarrow \infty} \frac{1}{2U} \int_{-U}^U Y(u) Y(u + \eta) du \quad (58)$$

a positive-definite tempered distribution. Because the map $Z \mapsto Y$ is an L^2 -isometry under $u = \Theta(t)$, this is equivalent to correlating Z against itself with respect to the u -shift η , which is a nonlinear (roughly logarithmic) shift in the original t -coordinate.

A.4 What does the picture look like?

Three objects merit mental plots.

The function $Y(u)$. An envelope-modulated real oscillation of zero mean. The envelope decays like $1/\sqrt{\log t}$ expressed in u -coordinates. The zero crossings are the images under Θ of the non-trivial Riemann zeros; the phase variable straightens out their spacing.

The autocorrelation $C(\eta)$. A sharp peak at $\eta = 0$, symmetric in η because Y is real, with oscillatory decay whose frequency content sits inside $[-1, 1]$.

The spectral measure S . A non-negative tempered measure strictly supported in $[-1, 1]$, symmetric about $\xi = 0$ after the even extension. Its shape inside $[-1, 1]$ encodes the distribution of zero spacings of Z in u -coordinates.

A.5 Locally L^2 and the growing variance

Y is locally L^2 , on compact subsets $[-U, U]$ of arbitrarily large length U . The band-limit established in Theorem 1 is a statement about the Fourier support of Y as a tempered distribution on $\mathbb{R}$. All uses of Paley–Wiener, Bochner–Schwartz, and Akhiezer in the main text are read in the tempered-distributional form, the framework for band-limited objects that are not globally L^2 .

The growing variance manifests in three compatible ways.

Envelope in u -coordinates. $Y(u)$ has an amplitude envelope that grows slowly as $|u| \rightarrow \infty$; the local mean-square $\bar{P}(U) := \frac{1}{2U} \int_{-U}^U Y(u)^2 du$ grows without bound as $U \rightarrow \infty$.

Autocorrelation as a tempered distribution. $C(\eta)$ is the windowed-limit positive-definite distribution of §A.3. Bochner–Schwartz identifies its Fourier transform with the non-negative tempered measure S , and Theorem 1 localizes S to $[-1, 1]$.

Spectral measure as a tempered measure on $[-1, 1]$. S is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$, of infinite total mass.

Representation at finite window. Pointwise values of the spectral density at interior $\xi_0 \in (-1, 1)$ are approached through windowed periodograms

$$P_U(\xi_0) := \frac{1}{2U} \left| \int_{-U}^U Y(u) e^{-i\xi_0 u} du \right|^2 \quad (59)$$

Any figure produced at finite U is a *representation* at window scale U .

A.6 Scale of the spectral representation at finite window

Fix $\xi \in (-1, 1)$ and let $P_U(\xi)$ be the windowed periodogram. Two features of the plot change with U : the overall vertical scale, and the local fluctuation amplitude around that scale. Both are controlled by the local mean power

$$\bar{P}(U) := \frac{1}{2U} \int_{-U}^U Y(u)^2 du = \frac{1}{2U} \int_{\Theta^{-1}(-U)}^{\Theta^{-1}(U)} Z(t)^2 dt \quad (60)$$

the second equality by the L^2 -isometry of the unitary inverse phase pullback. By Parseval in the tempered sense,

$$\int_{-\infty}^{\infty} P_U(\xi) d\xi = 2\pi \bar{P}(U) \quad (61)$$

Using $\Theta(t) \sim \frac{t}{2} \log(t/(2\pi e)) + ct$ and the Hardy–Littlewood moment $\int_0^T Z(t)^2 dt \sim T \log T$ ([11], §7.3),

$$\bar{P}(U) \sim \log(\Theta^{-1}(U)) \sim \log U + \log \log U + O(1) \quad (U \rightarrow \infty) \quad (62)$$

$P_U(\xi)$ fluctuates around $\bar{P}(U)$ on a scale comparable to $\bar{P}(U)$ itself, with wider fluctuations near the edges $|\xi| \rightarrow 1^-$.

A.7 The normalized spectral measure

At every finite U the periodogram P_U has finite total mass $2\pi\bar{P}(U)$ by Parseval. Dividing by this total mass produces the normalized spectral probability measure at scale U ,

$$\tilde{P}_U(\xi) := \frac{P_U(\xi)}{2\pi\bar{P}(U)}, \quad \int_{-\infty}^{\infty} \tilde{P}_U(\xi) d\xi = 1 \quad (63)$$

a probability density on $\mathbb{R}$ at every U . By Theorem 1 the probability mass of $\tilde{P}_U$ outside $[-1, 1]$ vanishes as $U \rightarrow \infty$, so the family is tight on $[-1, 1]$. By Helly–Prokhorov, $\tilde{P}_U$ converges weakly as $U \rightarrow \infty$ to a probability measure ν_∞ on $[-1, 1]$:

$$\int f d\tilde{P}_U \longrightarrow \int f d\nu_\infty \quad \text{for every } f \in C_b(\mathbb{R}) \quad (64)$$

ν_∞ is the normalized spectral density of Y : a probability measure of total mass 1 supported in $[-1, 1]$.

S and ν_∞ are distinct, both well-defined:

- S : non-negative tempered spectral measure on $[-1, 1]$, infinite total mass, delivered by Theorem 1 and Bochner–Schwartz ([9], Theorem 7.7).
- ν_∞ : probability measure on $[-1, 1]$, the weak-* limit of the normalized windowed periodograms $\tilde{P}_U$.

A.8 Cramér representation of the sample path

As a generalized second-order stationary process on $\mathbb{R}_u$ with non-negative tempered spectral measure S supported in $[-1, 1]$, Y admits a Cramér spectral representation of its sample path,

$$Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi) \quad (65)$$

where $d\Phi$ is a complex-valued orthogonal Borel measure on $[-1, 1]$ with covariance structure

$$\mathbb{E}[d\Phi(\xi) \overline{d\Phi(\xi')}] = \delta(\xi - \xi') dS(\xi) \quad (66)$$

The support condition $\text{supp } S \subseteq [-1, 1]$ of Theorem 1 localizes $d\Phi$ to the interval $[-1, 1]$. Reality of Y on $\mathbb{R}$ is equivalent to the Hermitian symmetry $d\Phi(-\xi) = \overline{d\Phi(\xi)}$, under which S is an even measure on $[-1, 1]$. Test functions f are paired with $d\Phi$ under the $L^2(S)$ isometry

$$\mathbb{E} \left| \int f(\xi) d\Phi(\xi) \right|^2 = \int |f(\xi)|^2 dS(\xi) \quad (67)$$

which requires $f \in L^2(S)$ and permits S to have infinite total mass. The Cramér representation exists at the level of S : it is the reconstruction theorem for the process. ν_∞ is the scale-invariant spectral shape of the same process.

Inverse representation of $d\Phi$. The orthogonal Borel measure $d\Phi$ is recovered from the sample path Y by the truncated Fourier inversion

$$\Phi(B) = \lim_{U \rightarrow \infty} \frac{1}{2\pi} \int_{-U}^U Y(u) \widehat{\mathcal{K}}_B(-u) du \quad (68)$$

for every Borel set $B \subseteq [-1, 1]$, the limit taken in $L^2(\Omega_P, \mathbb{P})$; here $\widehat{\mathcal{K}}_B(u) = \int_B e^{i\xi u} d\xi$. Equivalently, for every $f \in L^2(S)$,

$$\int_{-1}^1 f(\xi) d\Phi(\xi) = \lim_{U \rightarrow \infty} \frac{1}{2\pi} \int_{-U}^U Y(u) \check{f}(u) du, \quad \check{f}(u) = \int_{-1}^1 f(\xi) e^{-i\xi u} d\xi \quad (69)$$

again in $L^2(\Omega_P, \mathbb{P})$. Existence of the limit is the $L^2(S)$ isometry (67) together with the truncated Fourier inversion on Y .

Inserting $d\Phi$ into the forward representation. Fix admissible c and $t \in \mathbb{R}$. Apply (69) with test function $f(\xi) = e^{i\xi\Theta(t)}$; this f is continuous on $[-1, 1]$ hence in $L^2(S|_{[-1, 1]})$. The inner ξ -integral in the inverse formula is Lebesgue,

$$\check{f}(u) = \int_{-1}^1 e^{i\xi\Theta(t)} e^{-i\xi u} d\xi = \int_{-1}^1 e^{i\xi(\Theta(t)-u)} d\xi = \frac{2 \sin(\Theta(t) - u)}{\Theta(t) - u} \quad (70)$$

so that (69) reads

$$\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) = \lim_{U \rightarrow \infty} \frac{1}{\pi} \int_{-U}^U Y(u) \frac{\sin(\Theta(t) - u)}{\Theta(t) - u} du \quad (71)$$

in $L^2(\Omega_P, \mathbb{P})$. The passage from the nested (ξ, u) -form of (69) to the single u -integral (71) exchanges the order of ξ - and u -integration. At every finite U this is Fubini on the product measure $d\xi \otimes du$ on $[-1, 1] \times [-U, U]$: the integrand satisfies

$$|Y(u) e^{i\xi(\Theta(t)-u)}| = |Y(u)| \quad (72)$$

and

$$\int_{-U}^U \int_{-1}^1 |Y(u)| d\xi du = 2 \int_{-U}^U |Y(u)| du < \infty \quad (73)$$

since Y is locally L^2 on $\mathbb{R}$ hence locally L^1 on the compact $[-U, U]$; $[-1, 1]$ has finite Lebesgue measure. Fubini therefore applies: the double integral factors in either order, with the same value,

$$\int_{-U}^U Y(u) \left(\int_{-1}^1 e^{i\xi(\Theta(t)-u)} d\xi \right) du = \int_{-1}^1 \left(\int_{-U}^U Y(u) e^{i\xi(\Theta(t)-u)} du \right) d\xi \quad (74)$$

and the $L^2(\Omega_P, \mathbb{P})$ -limit $U \rightarrow \infty$ on both sides exists, with the limit on the right equal to $\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi)$ by truncated Fourier inversion for Y . The bound (72) is independent of c , and the only appearance of c in (74) is through the real translate $\Theta(t) = \theta(t) + ct$ in the phase; Fubini's hypothesis is therefore satisfied for every admissible c .

Time-warped representation of Z . The limit on the right of (71) is the sinc-kernel sampling of Y at the point $\Theta(t) \in \mathbb{R}$. Since Y is band-limited to $[-1, 1]$ by Theorem 1 and $\Theta(t) \in \mathbb{R}$, Paley–Wiener sampling gives

$$\lim_{U \rightarrow \infty} \frac{1}{\pi} \int_{-U}^U Y(u) \frac{\sin(\Theta(t) - u)}{\Theta(t) - u} du = Y(\Theta(t)) \quad (75)$$

in $L^2(\Omega_P, \mathbb{P})$. Combining (71), (75), and the real-line identity (49) ($Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$),

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)) = \sqrt{\Theta'(t)} \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \quad (76)$$

for every admissible c and every $t \in \mathbb{R}$. This is the unitary time-change of the Cramér representation: the same orthogonal Borel measure $d\Phi$ that represents the stationary sample path Y represents Z after the reparametrisation $u \mapsto \Theta(t)$ and the half-Jacobian amplitude $\sqrt{\Theta'(t)}$. The representation (76) depends on c only through $\Theta = \theta + c \cdot \text{id}$; the orthogonal Borel measure $d\Phi$ itself is a spectral object of Y , independent of c .

Covariance of Z . From (76), the covariance of Z at $t, s \in \mathbb{R}$ is

$$R_Z(t, s) := \mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\Theta'(t) \Theta'(s)} \int_{-1}^1 e^{i\xi(\Theta(t) - \Theta(s))} dS(\xi) \quad (77)$$

Derivation of (77) from (76). Fix $t, s \in \mathbb{R}$. Expanding the product and pulling the amplitudes outside the expectation,

$$\mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\Theta'(t) \Theta'(s)} \mathbb{E} \left[\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \overline{\int_{-1}^1 e^{i\xi'\Theta(s)} d\Phi(\xi')} \right] \quad (78)$$

The two Wiener integrals are each elements of $L^2(\Omega_P, \mathbb{P})$ with $L^2(S)$ -norms equal to $\|e^{i\cdot\Theta(t)}\|_{L^2(S)} = S([-1, 1])^{1/2}$ and $\|e^{i\cdot\Theta(s)}\|_{L^2(S)} = S([-1, 1])^{1/2}$, which are finite whenever the integrals are taken against a compactly supported test function — here the restriction of $e^{i\cdot\Theta(t)}$ and $e^{i\cdot\Theta(s)}$ to the compact set $[-1, 1]$. The Wiener–Itô product rule (the bilinear form of (67)) gives

$$\mathbb{E} \left[\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \overline{\int_{-1}^1 e^{i\xi'\Theta(s)} d\Phi(\xi')} \right] = \int_{-1}^1 e^{i\xi\Theta(t)} \overline{e^{i\xi\Theta(s)}} dS(\xi) \quad (79)$$

Combining (78) and (79) gives (77).

The exchange of the two ξ -integrals with the expectation in (79) is an instance of the bilinear Itô isometry and not an application of Fubini directly. The Fubini justification of that bilinear isometry is standard: for simple functions $f = \sum_k a_k \mathbb{1}_{B_k}$ with $B_k \subseteq [-1, 1]$ pairwise disjoint and Borel, and $g = \sum_\ell b_\ell \mathbb{1}_{C_\ell}$ similarly, orthogonality of the increments $\{\Phi(B_k)\}$ gives

$$\mathbb{E} \left[\int f d\Phi \overline{\int g d\Phi} \right] = \sum_k \sum_\ell a_k \overline{b_\ell} \mathbb{E}[\Phi(B_k) \overline{\Phi(C_\ell)}] = \sum_k a_k \overline{b_k} S(B_k \cap C_k) = \int f \overline{g} dS \quad (80)$$

where in the middle equality we used $\mathbb{E}[\Phi(B_k) \overline{\Phi(C_\ell)}] = S(B_k \cap C_\ell)$ (the integrated form of (66)) and the fact that for simple functions the double sum is finite so Fubini applies pointwise. Bounded-continuous functions on $[-1, 1]$ are $L^2(S)$ -limits of simple functions (standard, since $S([-1, 1]) < \infty$)

if one replaces S by S -on- $[-1, 1]$, which is finite because Theorem 1 supports S in $[-1, 1]$ but does not bound its total mass; for the identity (79) the tests $e^{i\cdot\Theta(t)}$ and $e^{i\cdot\Theta(s)}$ are bounded on the compact set $[-1, 1]$, so they are in $L^2(S|_{[-1, 1]})$ even when S itself is infinite on $\mathbb{R}$, and the bilinear form in (79) extends continuously in both arguments by the $L^2(S)$ isometry (67), yielding the claim. No assumption on c is used beyond admissibility, which is the standing hypothesis on Θ .

The exchange in (76) itself — the equality $\sqrt{\Theta'(t)}Y(\Theta(t)) = \sqrt{\Theta'(t)}\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi)$ — is the pointwise substitution $u \mapsto \Theta(t)$ in (65), valid because $\Theta(t) \in \mathbb{R}$ for $t \in \mathbb{R}$ and the sample path $Y(\Theta(t))$ is defined by the Cramér representation evaluated at the point $\Theta(t)$. No integral exchange is required: the inner Wiener integral is the same element of $L^2(\Omega_P, \mathbb{P})$, with the phase $e^{i\xi u}$ evaluated at $u = \Theta(t)$. $\square$

The identity (77) is the time-warped autocovariance of the stationary underlying process. Non-stationarity of Z is expressed entirely through the pair $(\Theta'(t)\Theta'(s))^{1/2}$ and the argument $\Theta(t) - \Theta(s)$: the frequency content is the same S supported in $[-1, 1]$; the time dependence is absorbed into the biholomorphic reparametrisation Θ and the half-Jacobian amplitude.

A.9 Variance structure function

At coincident times $s = t$ the covariance (77) reduces to the variance of $Z(t)$, and the ξ -integral in (77) loses its oscillatory factor. Write

$$\sigma_Y^2 := S([-1, 1]) = \int_{-1}^1 dS(\xi) \quad (81)$$

for the total mass of S restricted to $[-1, 1]$; this quantity is finite when Y is second-order on $\mathbb{R}$, which is the hypothesis of the Cramér representation (65). Setting $s = t$ in (77) then gives $\Theta(t) - \Theta(s) = 0$, the exponential equals 1, and

$$\text{Var } Z(t) = R_Z(t, t) = \Theta'(t) \int_{-1}^1 dS(\xi) = \sigma_Y^2 \Theta'(t) \quad (82)$$

for every $t \in \mathbb{R}$. The variance of Z is the total spectral mass of Y amplified by the instantaneous phase rate $\Theta'(t)$; it is the diagonal restriction of the covariance, with no dependence on the off-diagonal phase.

Pinning of σ_Y^2 by the Hardy–Littlewood second moment. The value $\sigma_Y^2 = 2$ is not a free calibration; it is pinned as a theorem by the Hardy–Littlewood second moment of ζ . Integrating (82) against dt on $[0, T]$,

$$\int_0^T \text{Var } Z(t) dt = \sigma_Y^2 \int_0^T \Theta'(t) dt = \sigma_Y^2 [\Theta(T) - \Theta(0)]. \quad (83)$$

On the left, using $|Z(t)| = |\zeta(\frac{1}{2} + it)|$ on $\mathbb{R}$ (from $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ with $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$),

$$\int_0^T \text{Var } Z(t) dt = \int_0^T \mathbb{E} |\zeta(\frac{1}{2} + it)|^2 dt = \int_0^T |\zeta(\frac{1}{2} + it)|^2 dt, \quad (84)$$

the outer expectation being trivial since $|\zeta(\frac{1}{2} + it)|^2$ is not a chance quantity. On the right, $\Theta(T) = \theta(T) + cT = \frac{T}{2} \log(T/(2\pi)) - \frac{T}{2} + cT + O(1)$. Hardy–Littlewood ([11], §7.3) gives $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log T - (1 + \log 2\pi - 2\gamma)T + O(T^{1/2} \log T)$. Dividing both (83) and the Hardy–Littlewood identity

by $T \log T$ and taking $T \rightarrow \infty$ equates the two leading coefficients: the right side of (83) contributes $\sigma_Y^2 \cdot \frac{1}{2}$ and Hardy–Littlewood contributes 1, forcing

$$\sigma_Y^2 = 2. \quad (85)$$

No degree of freedom remains; $\sigma_Y^2 = 2$ is a theorem derived from Hardy–Littlewood, not a convention. With $\sigma_Y^2 = 2$, (82) becomes the exact identity

$$\text{Var } Z(t) = 2 \Theta'(t), \quad (86)$$

with $\Theta'(t) = \theta'(t) + c = \frac{1}{2} \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log \pi + c$ by (3). No asymptotic expansion of θ' enters: (86) holds for every $t \in \mathbb{R}$ as an identity between $\text{Var } Z$ and the digamma function.

A.10 Sample-path autocorrelation under the unitary time change

The stationary process Y carries a sample-path autocorrelation, and the unitary time change expresses it directly on Z . For any window $[A, B] \subset \mathbb{R}_u$ in the u -variable and any lag $\eta \in \mathbb{R}$,

$$C_Y(\eta; A, B) := \int_A^{B-\eta} Y(u) \overline{Y(u+\eta)} du \quad (87)$$

is the unnormalized sample autocorrelation of Y on $[A, B]$ at lag η . Apply the substitution $u = \Theta(\tau)$, $du = \Theta'(\tau) d\tau$, using $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ from (8). The pair $(u, u + \eta)$ becomes $(\Theta(\tau), \Theta(\tau) + \eta)$, and the second argument is the image of $\Theta^{-1}(\Theta(\tau) + \eta)$ under Θ . Writing $\tau'(\tau, \eta) := \Theta^{-1}(\Theta(\tau) + \eta)$ for the lagged time in t -coordinates,

$$C_Y(\eta; A, B) = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B-\eta)} Z(\tau) \overline{Z(\tau'(\tau, \eta))} \sqrt{\frac{\Theta'(\tau)}{\Theta'(\tau'(\tau, \eta))}} d\tau \quad (88)$$

where the integrand combines the two half-Jacobian factors $1/\sqrt{\Theta'(\tau)} \cdot 1/\sqrt{\Theta'(\tau'(\tau, \eta))}$ from the two evaluations of Y with the single Jacobian $\Theta'(\tau)$ from $du = \Theta'(\tau) d\tau$, giving the asymmetric ratio $\sqrt{\Theta'(\tau)/\Theta'(\tau'(\tau, \eta))}$. Equation (88) is the t -variable expression of the stationary sample-path autocorrelation of Y at lag η : translation-invariant on the u -axis becomes non-translation-invariant on the t -axis precisely through the asymmetric ratio and the Θ^{-1} -images of the window endpoints. At zero lag, $\tau'(\tau, 0) = \tau$, the ratio is 1, and

$$C_Y(0; A, B) = \int_A^B |Y(u)|^2 du = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B)} |Z(\tau)|^2 d\tau = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B)} \left| \zeta\left(\frac{1}{2} + i\tau\right) \right|^2 d\tau \quad (89)$$

using $|Z(\tau)| = |\zeta(\frac{1}{2} + i\tau)|$ on the real line. The right-hand side is the integrated second moment of ζ on the critical line over the window $[\Theta^{-1}(A), \Theta^{-1}(B)]$; as the window expands, it returns the Hardy–Littlewood asymptotic $T \log T$ and couples directly to (82) through $\sigma_Y^2 = 2$. Identity (88) is independent of c only up to the c -dependence of Θ : changing c rescales the window in t -coordinates through Θ^{-1} but does not change the integrated identity (89).

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